LTSpice Notch Quality Report

CROSS COUPLED10

Auto-detected swept parameters and quality-factor (Q = f0 / BW @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 2

Total simulation steps   : 33

`V(act+,act-)`

swept parameters: Cp, Itail

`V(pas+)`

swept parameters: Cp, Itail

Report contents (per signal column):

1. All curves — Bode plot (magnitude + phase)
2. Magnitude grouped by each swept parameter (one subplot per fixed value)
3. Phase grouped by each swept parameter (one subplot per fixed value)
4. Quality-factor table for that column

End of report (extra analyses):

A. Cross-signal comparison (act vs pas, if both present)

B. Q vs each swept parameter (one line per other-param combo)

C. BW vs each swept parameter

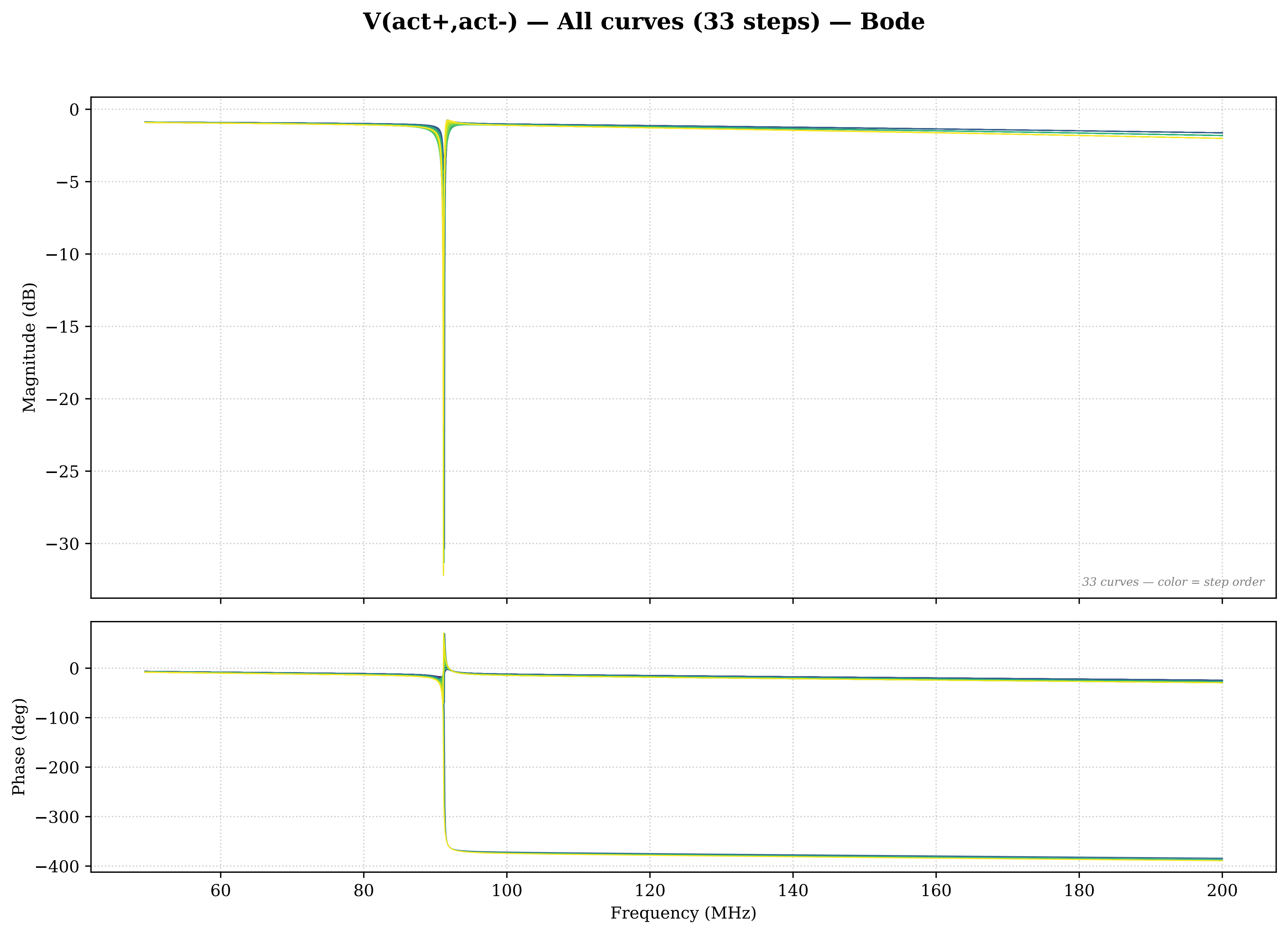
D. Q heatmap for V(act): rows × columns swept params, colored

E. Combined quality-factor table across all signals

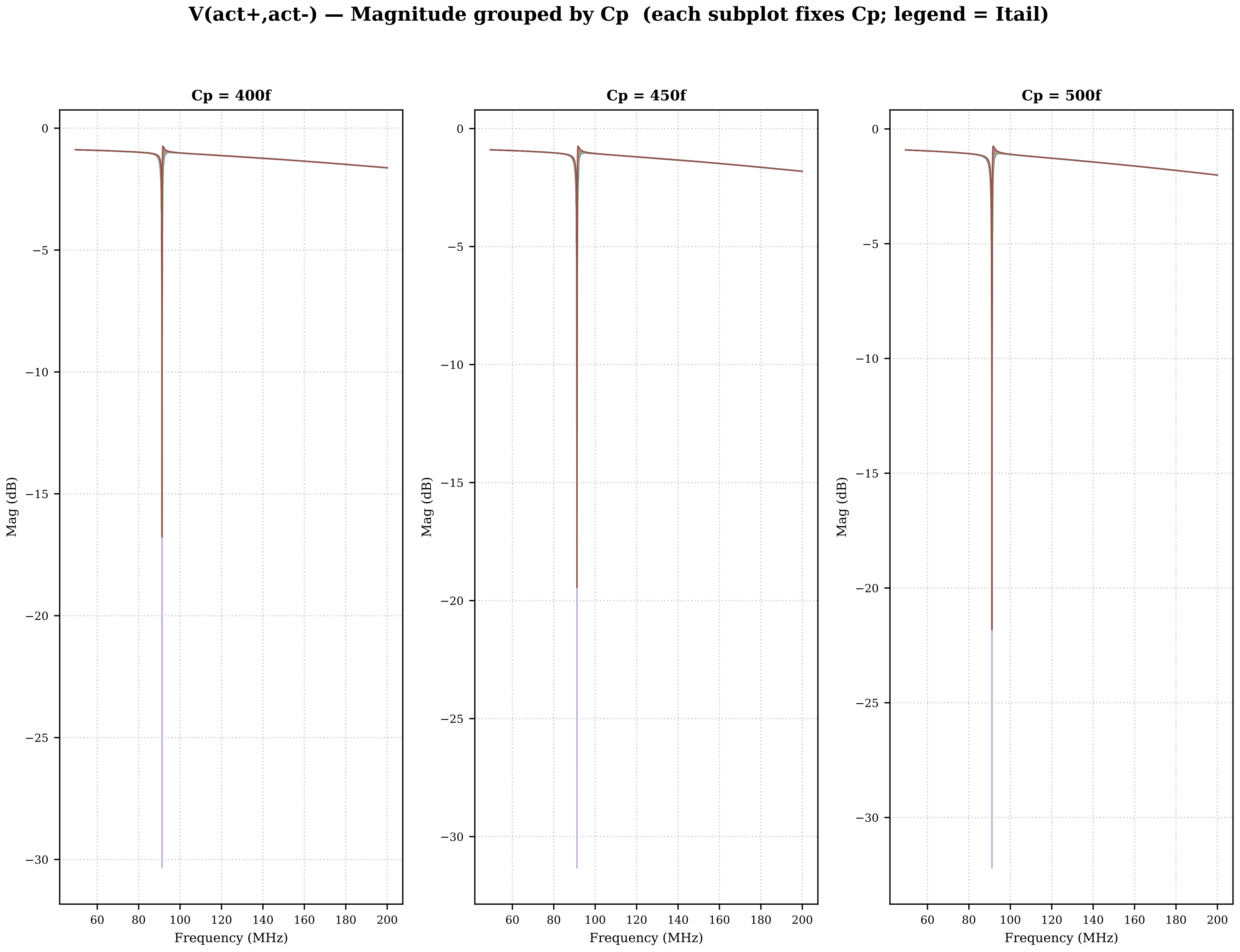
Signal: V(act+,act-)

Swept parameters: Cp, Itail

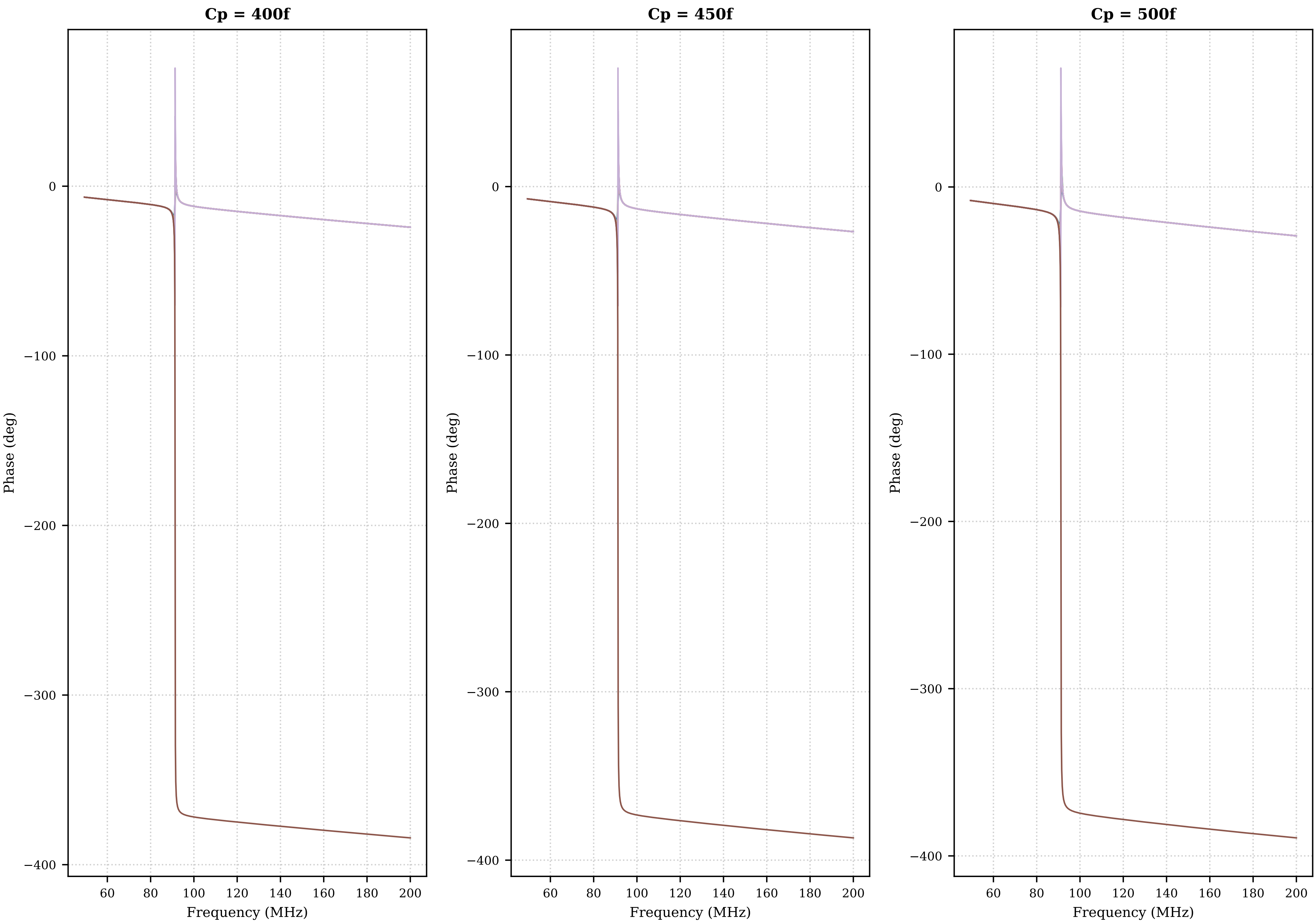
Steps: 33



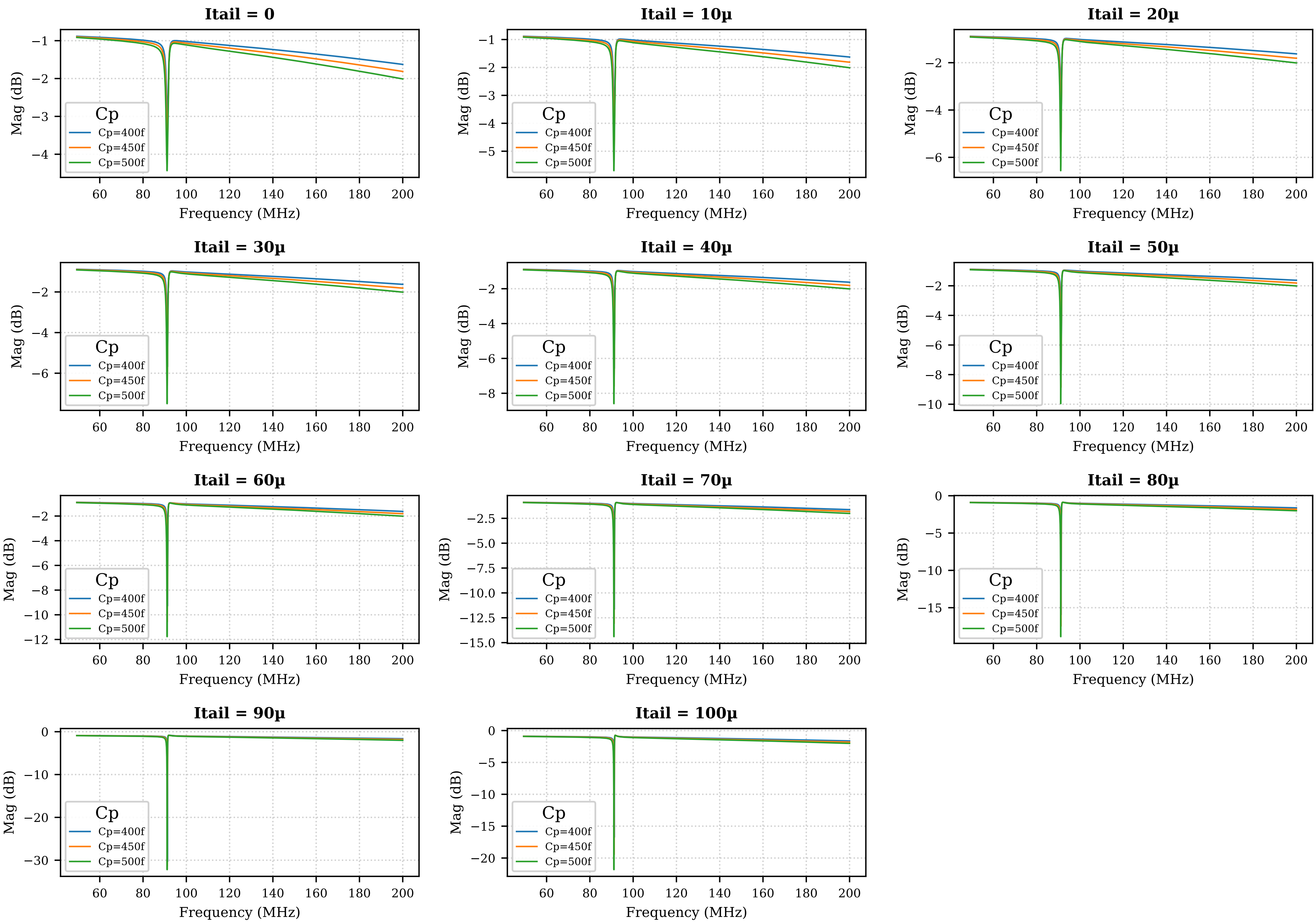
V(act+,act-) — All curves (33 steps) — Bode
Magnitude (dB)
0
−5
−10
−15
−20
−25
−30
33 curves — color = step order
Phase (deg)
0
−100
−200
−300
−400
60
80
100
120
140
160
180
200
Frequency (MHz)

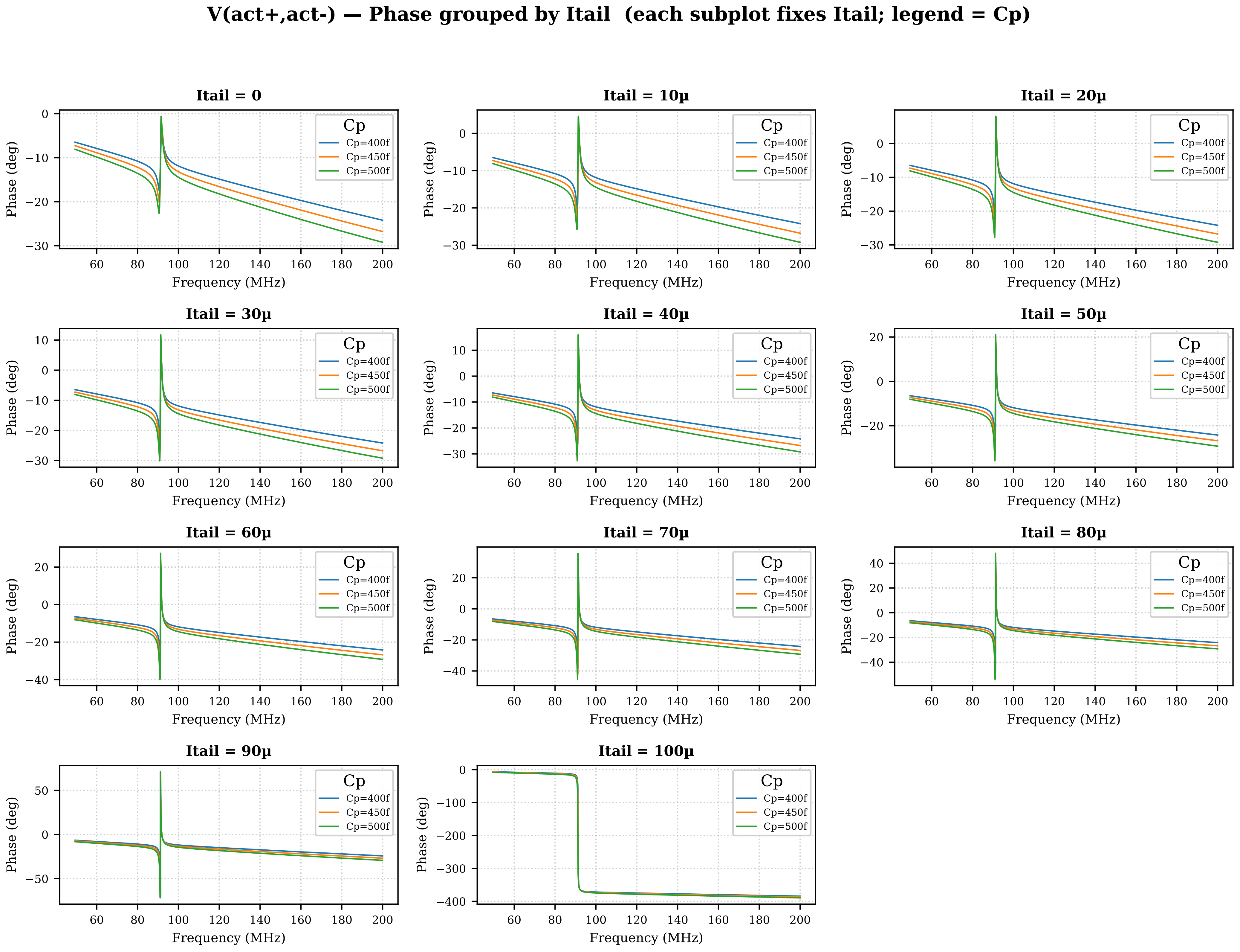

V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)
Cp = 400f
Cp = 450f
Cp = 500f
Mag (dB)
Frequency (MHz)
0
−5
−10
−15
−20
−25
−30
60
80
100
120
140
160
180
200

V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)



V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)




V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)
Itail = 0
Itail = 10μ
Itail = 20μ
Itail = 30μ
Itail = 40μ
Itail = 50μ
Itail = 60μ
Itail = 70μ
Itail = 80μ
Itail = 90μ
Itail = 100μ
Phase (deg)
Frequency (MHz)
Cp
Cp=400f
Cp=450f
Cp=500f

V(act+,act-) — Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	400f	0	91.25	-1.221	2.103			notch shallower than 3 d
V(act+,act-)	400f	10µ	91.26	-1.221	3.051	8.04e+04	1135	
V(act+,act-)	400f	20µ	91.26	-1.221	3.722	2.507e+05	364	
V(act+,act-)	400f	30µ	91.27	-1.221	4.466	3.002e+05	304	
V(act+,act-)	400f	40µ	91.27	-1.221	5.36	3.199e+05	285.3	
V(act+,act-)	400f	50µ	91.27	-1.221	6.506	3.238e+05	281.8	
V(act+,act-)	400f	60µ	91.27	-1.221	8.077	3.172e+05	287.8	
V(act+,act-)	400f	70µ	91.28	-1.221	10.45	3.018e+05	302.4	
V(act+,act-)	400f	80µ	91.28	-1.221	14.75	2.784e+05	327.8	
V(act+,act-)	400f	90µ	91.28	-1.221	29.13	2.461e+05	370.8	
V(act+,act-)	400f	100µ	91.28	-1.221	15.55	2.021e+05	451.7	
V(act+,act-)	450f	0	91.17	-1.316	2.553			notch shallower than 3 d
V(act+,act-)	450f	10µ	91.18	-1.315	3.662	2.992e+05	304.7	
V(act+,act-)	450f	20µ	91.19	-1.315	4.435	3.68e+05	247.8	
V(act+,act-)	450f	30µ	91.19	-1.315	5.279	3.932e+05	231.9	
V(act+,act-)	450f	40µ	91.2	-1.315	6.281	4.001e+05	227.9	
V(act+,act-)	450f	50µ	91.2	-1.315	7.546	3.959e+05	230.3	
V(act+,act-)	450f	60µ	91.2	-1.315	9.249	3.836e+05	237.7	
V(act+,act-)	450f	70µ	91.2	-1.315	11.77	3.645e+05	250.2	
V(act+,act-)	450f	80µ	91.2	-1.315	16.17	3.388e+05	269.2	
V(act+,act-)	450f	90µ	91.2	-1.315	30.02	3.06e+05	298.1	
V(act+,act-)	450f	100µ	91.2	-1.315	18.13	2.644e+05	344.9	
V(act+,act-)	500f	0	91.09	-1.417	3.017	6.909e+04	1319	
V(act+,act-)	500f	10µ	91.1	-1.417	4.28	4.307e+05	211.5	
V(act+,act-)	500f	20µ	91.11	-1.417	5.147	4.694e+05	194.1	
V(act+,act-)	500f	30µ	91.12	-1.417	6.082	4.809e+05	189.5	
V(act+,act-)	500f	40µ	91.12	-1.417	7.18	4.793e+05	190.1	
V(act+,act-)	500f	50µ	91.12	-1.417	8.545	4.693e+05	194.2	
V(act+,act-)	500f	60µ	91.12	-1.417	10.36	4.53e+05	201.2	
V(act+,act-)	500f	70µ	91.13	-1.417	12.98	4.311e+05	211.4	
V(act+,act-)	500f	80µ	91.13	-1.417	17.46	4.038e+05	225.7	
V(act+,act-)	500f	90µ	91.13	-1.417	30.78	3.709e+05	245.7	

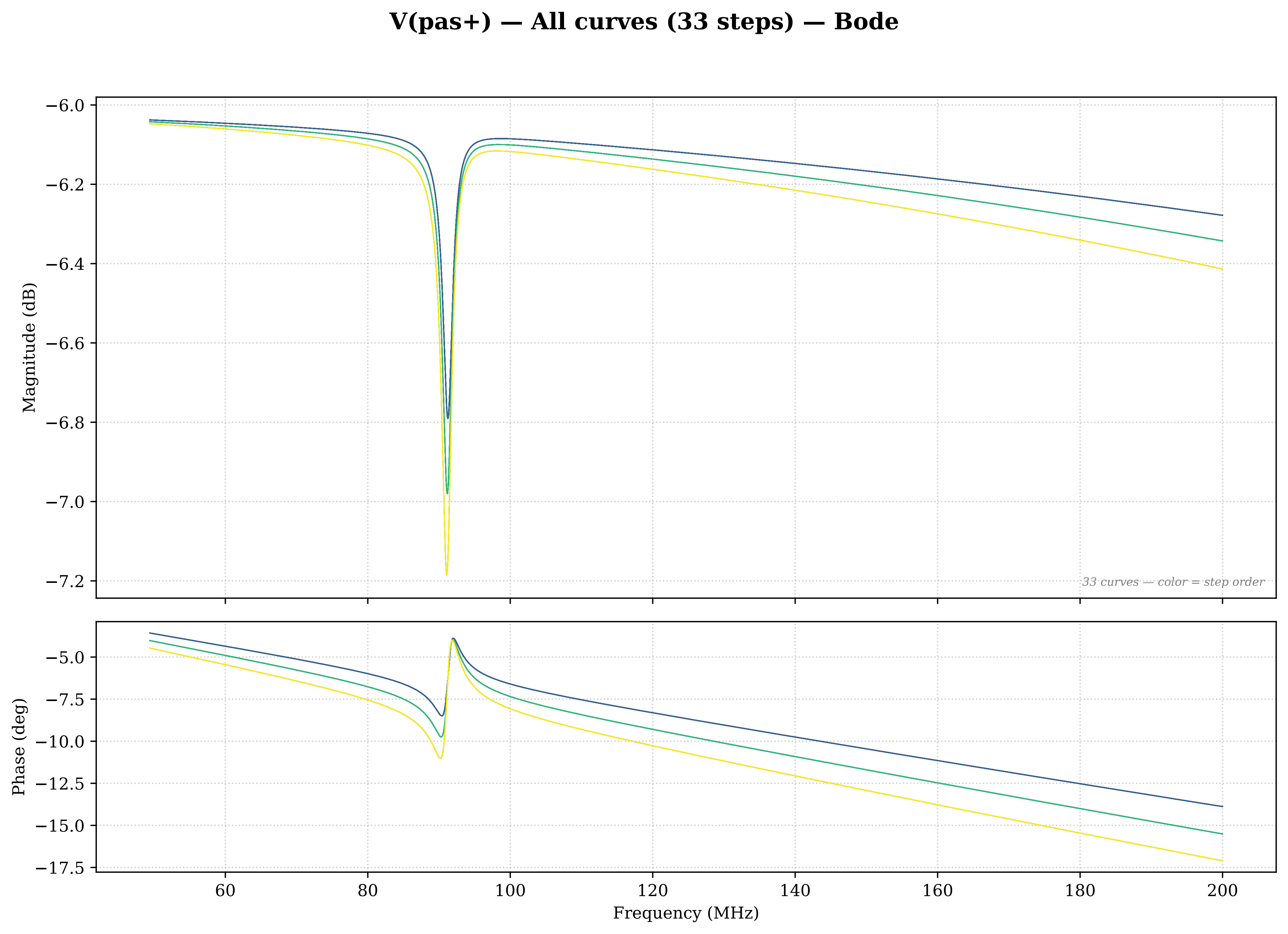
V(act+,act-) — Quality factor table (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	500f	100µ	91.13	-1.417	20.41	3.312e+05	275.1	

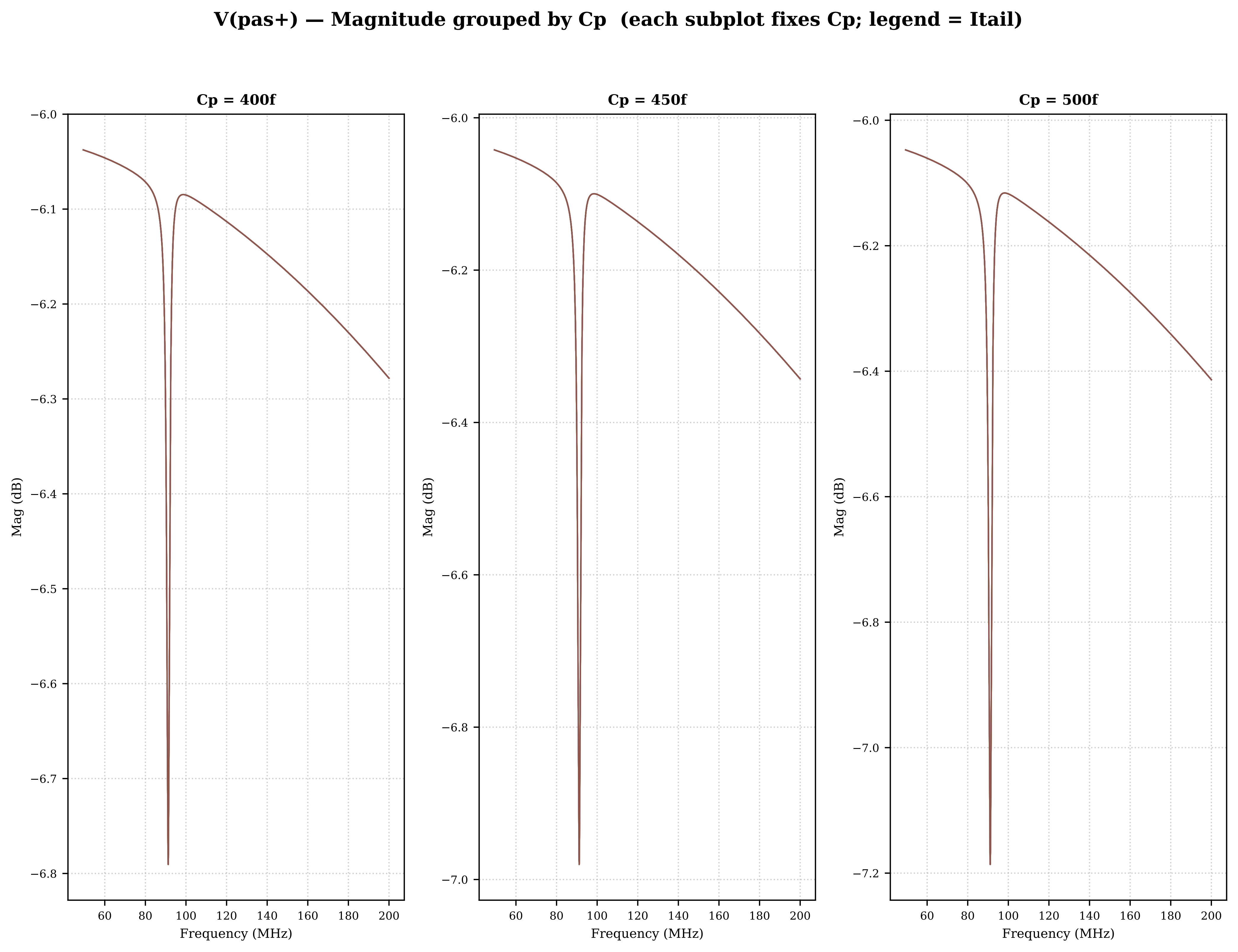
Signal: V(pas+)

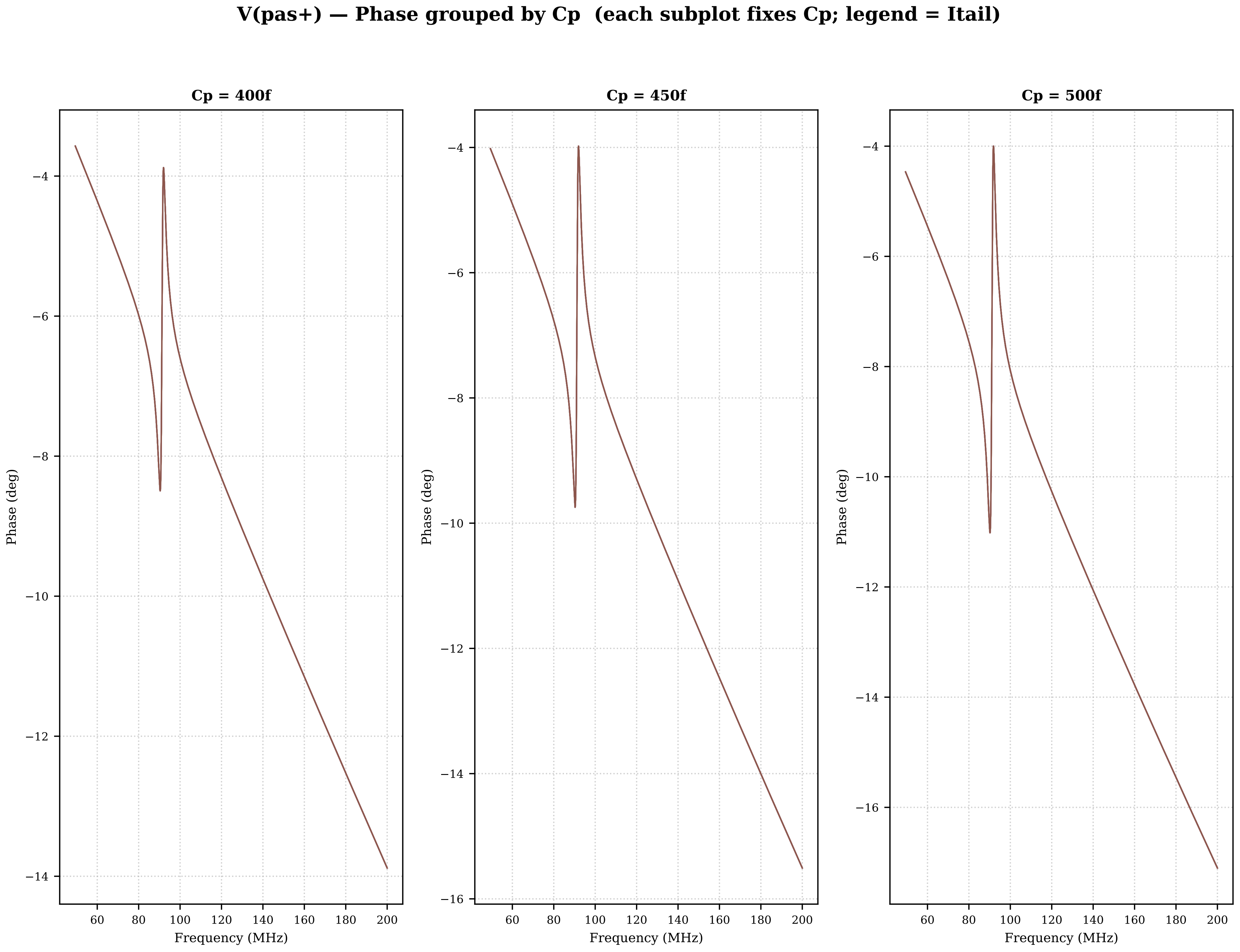
Swept parameters: Cp, Itail

Steps: 33

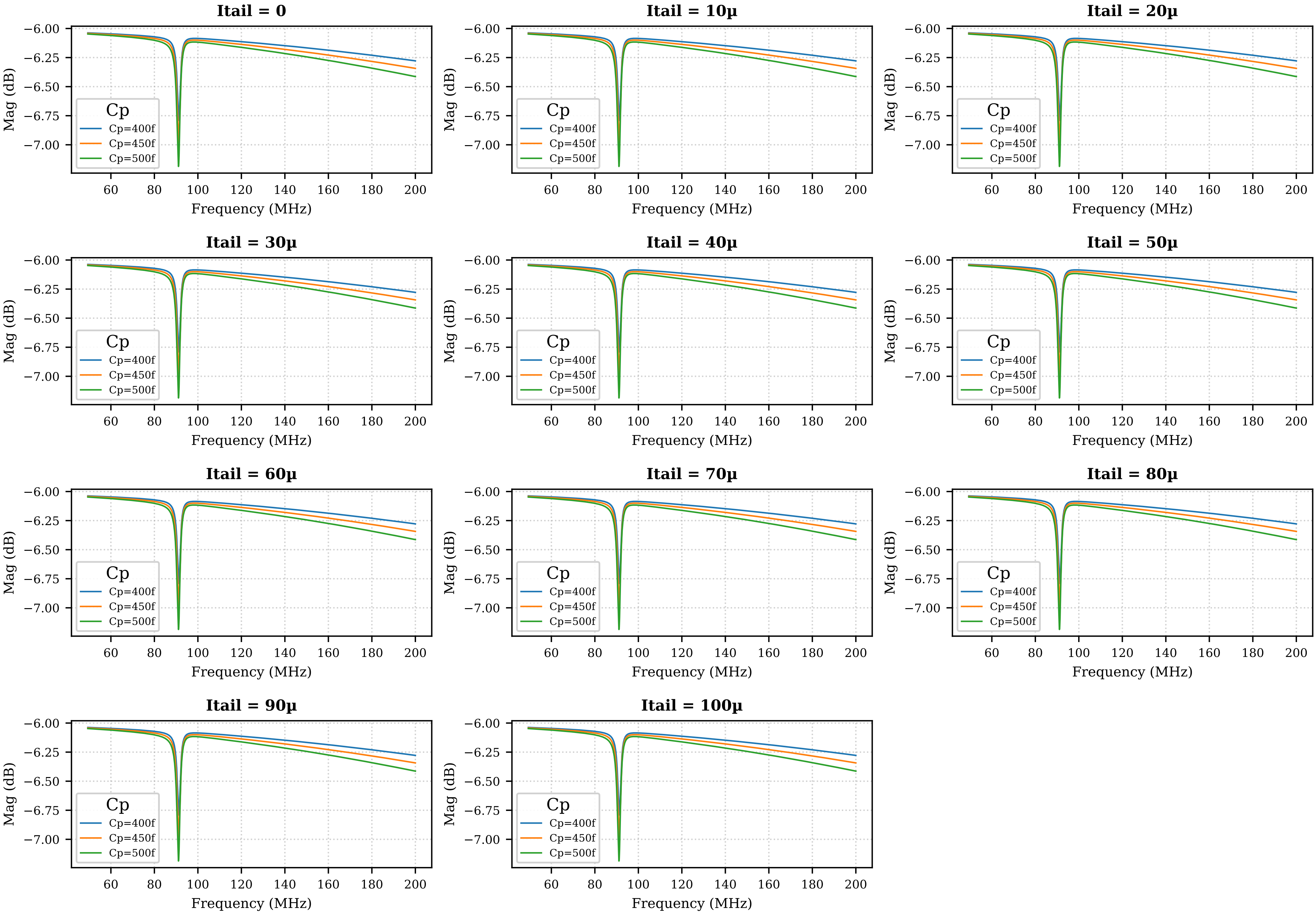


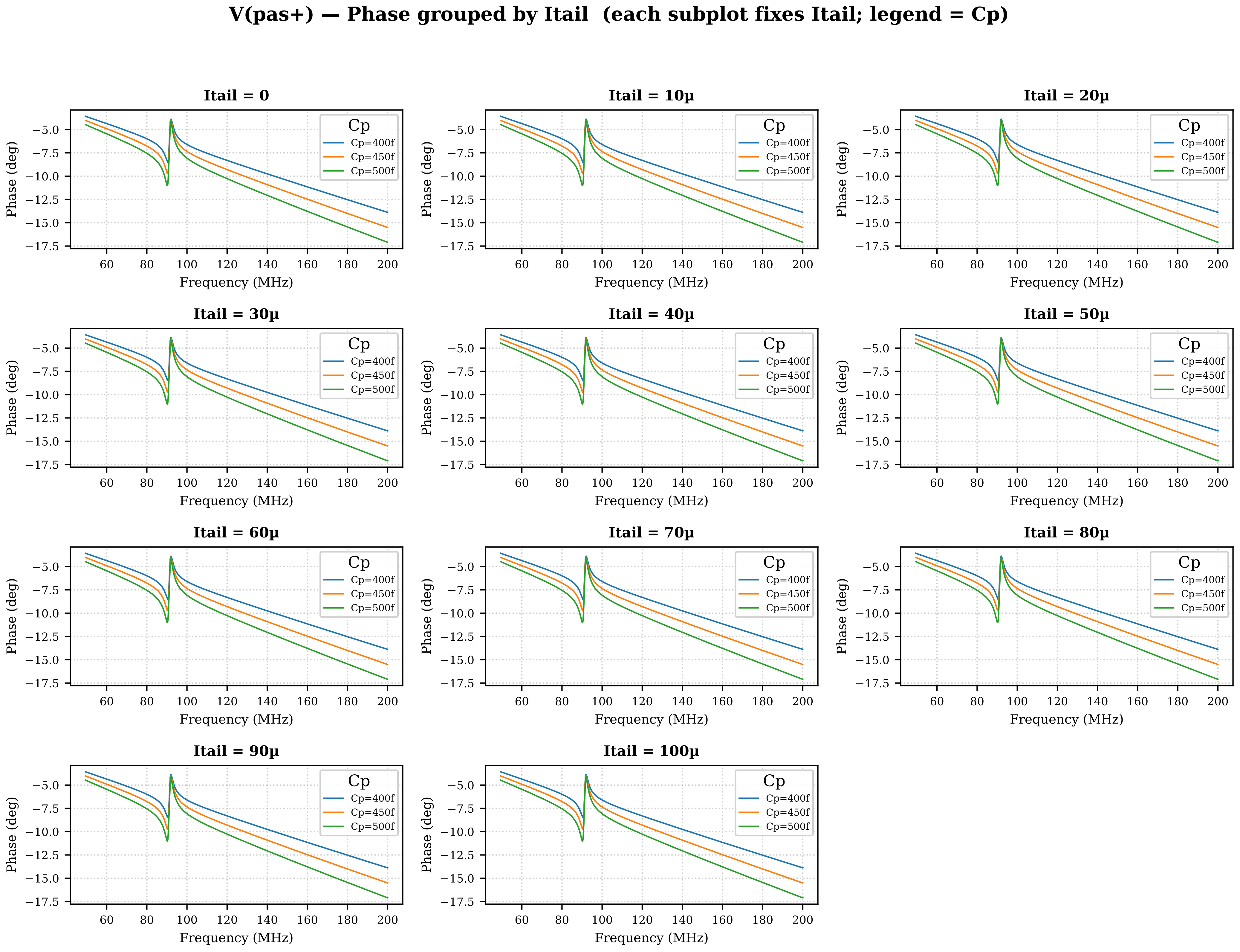
V(pas+) — All curves (33 steps) — Bode
Magnitude (dB)
−6.0
−6.2
−6.4
−6.6
−6.8
−7.0
−7.2
33 curves — color = step order
Phase (deg)
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
60
80
100
120
140
160
180
200
Frequency (MHz)


V(pas+) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)
Cp = 400f
Cp = 450f
Cp = 500f
Mag (dB)
Frequency (MHz)
60
80
100
120
140
160
180
200
−6.0
−6.1
−6.2
−6.3
−6.4
−6.5
−6.6
−6.7
−6.8
−7.0
−7.2


V(pas+) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)
Cp = 400f
Cp = 450f
Cp = 500f
Phase (deg)
Frequency (MHz)

V(pas+) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)




V(pas+) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)
Itail = 0
Itail = 10µ
Itail = 20µ
Itail = 30µ
Itail = 40µ
Itail = 50µ
Itail = 60µ
Itail = 70µ
Itail = 80µ
Itail = 90µ
Itail = 100µ
Cp
Cp=400f
Cp=450f
Cp=500f
Phase (deg)
Frequency (MHz)
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
60
80
100
120
140
160
180
200

V(pas+) — Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	400f	0	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	10µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	20µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	30µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	40µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	50µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	60µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	70µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	80µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	90µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	100µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	450f	0	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	10µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	20µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	30µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	40µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	50µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	60µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	70µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	80µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	90µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	100µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	500f	0	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	10µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	20µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	30µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	40µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	50µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	60µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	70µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	80µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	90µ	91.08	-6.213	0.9736			notch shallower than 3 d

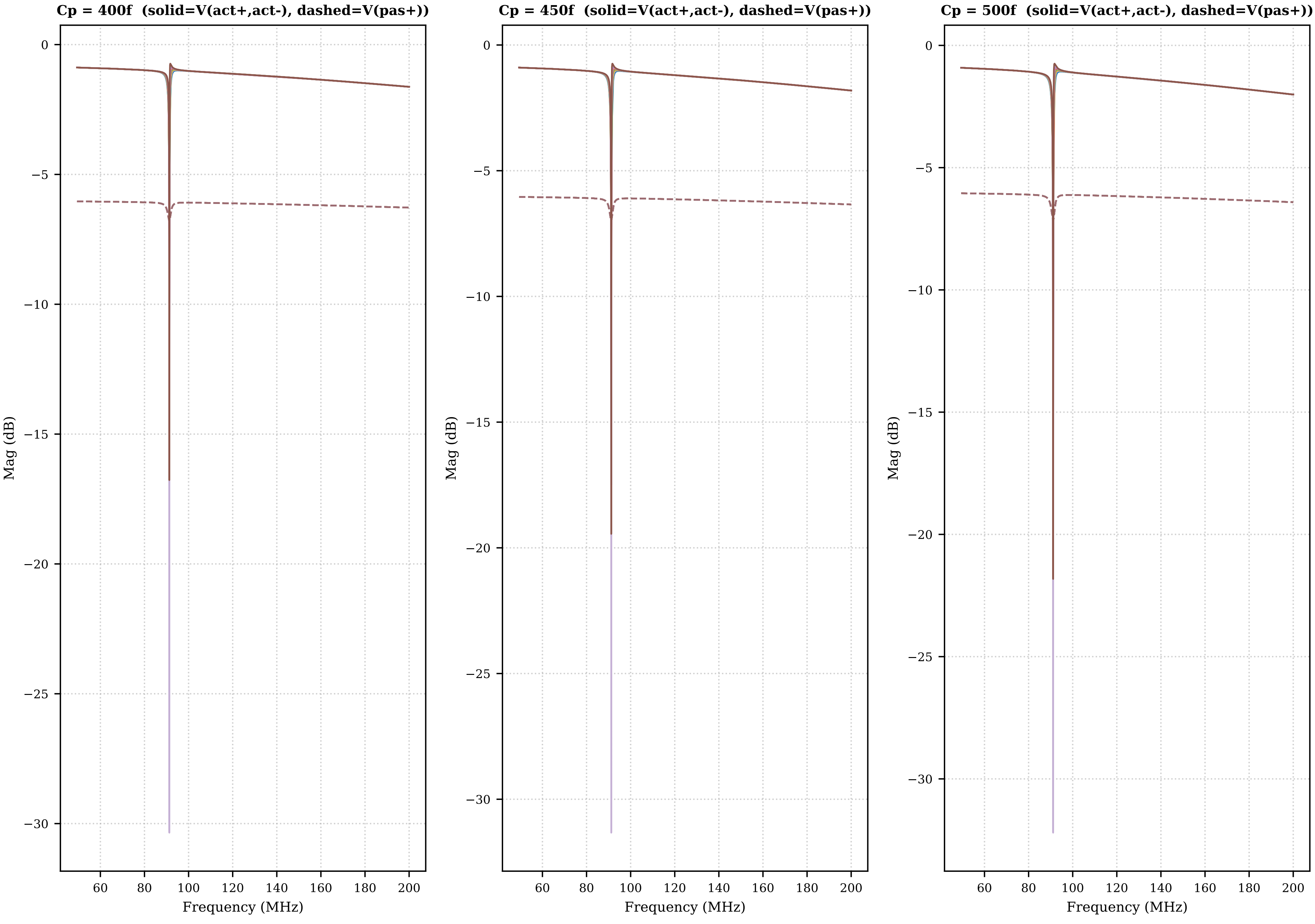
V(pas+) — Quality factor table (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	500f	100µ	91.08	-6.213	0.9736			notch shallower than 3 d

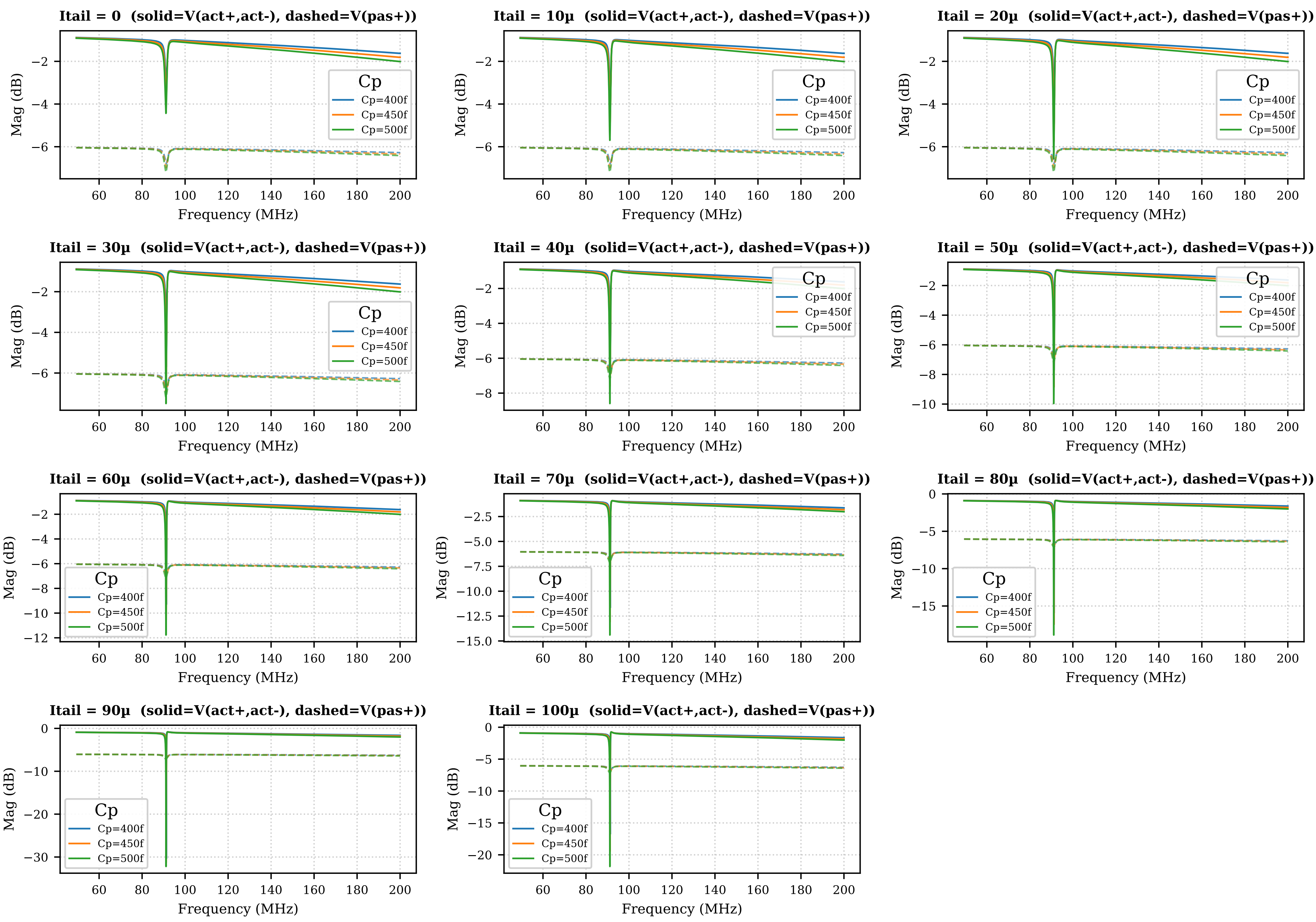
A. Active vs Passive comparison

Solid = V(act+,act-)   Dashed = V(pas+)

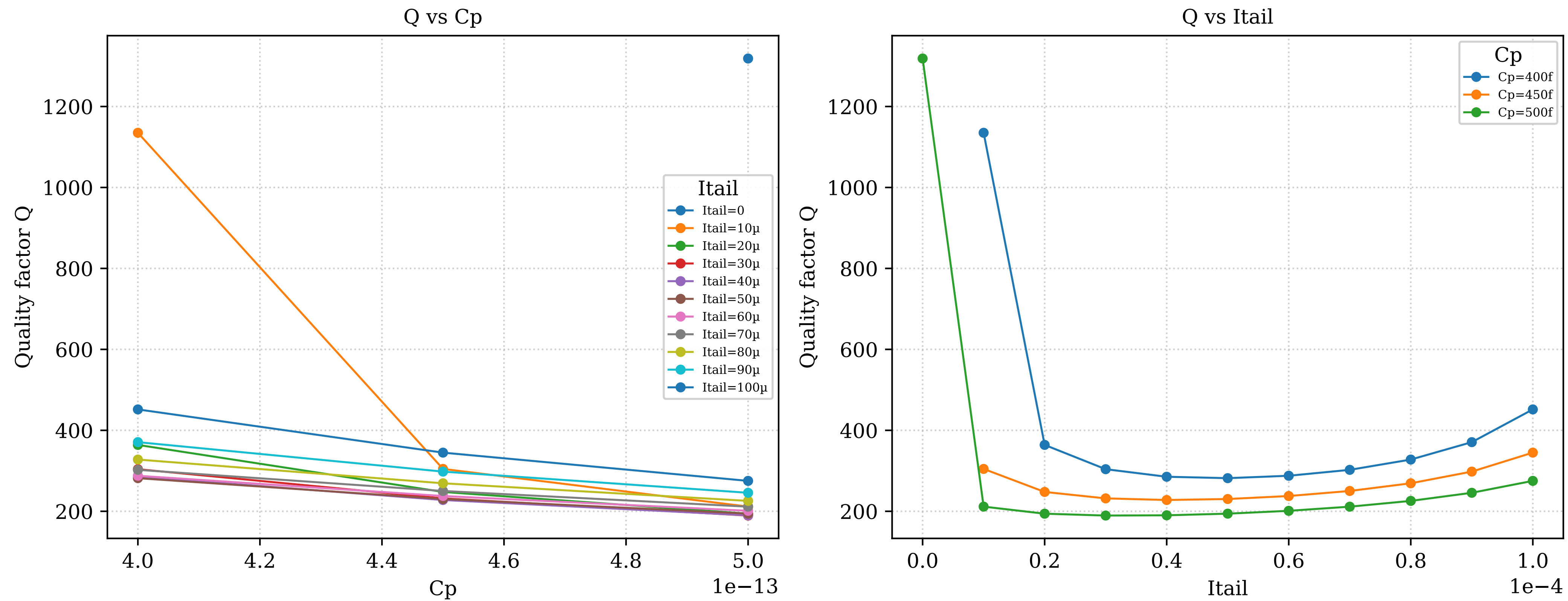
A. V(act+,act-) vs V(pas+) — grouped by Cp (legend = Itail)



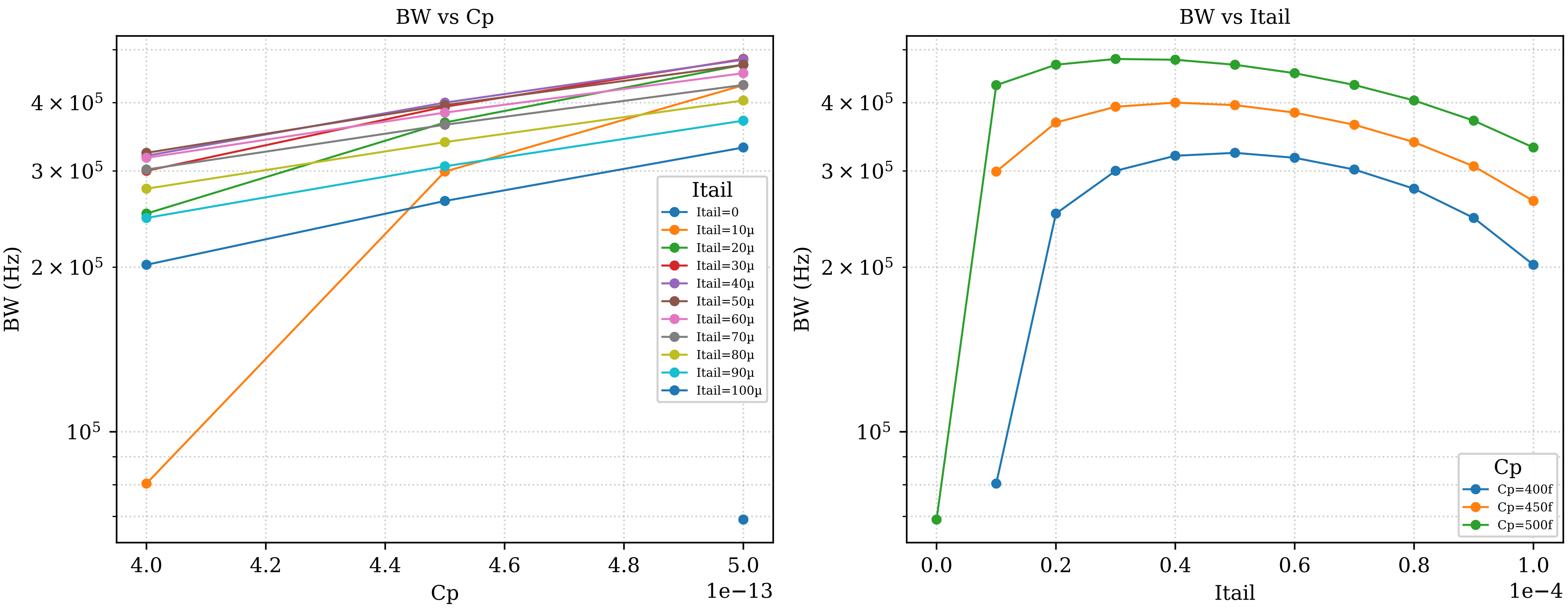
A. V(act+,act-) vs V(pas+) — grouped by Itail (legend = Cp)



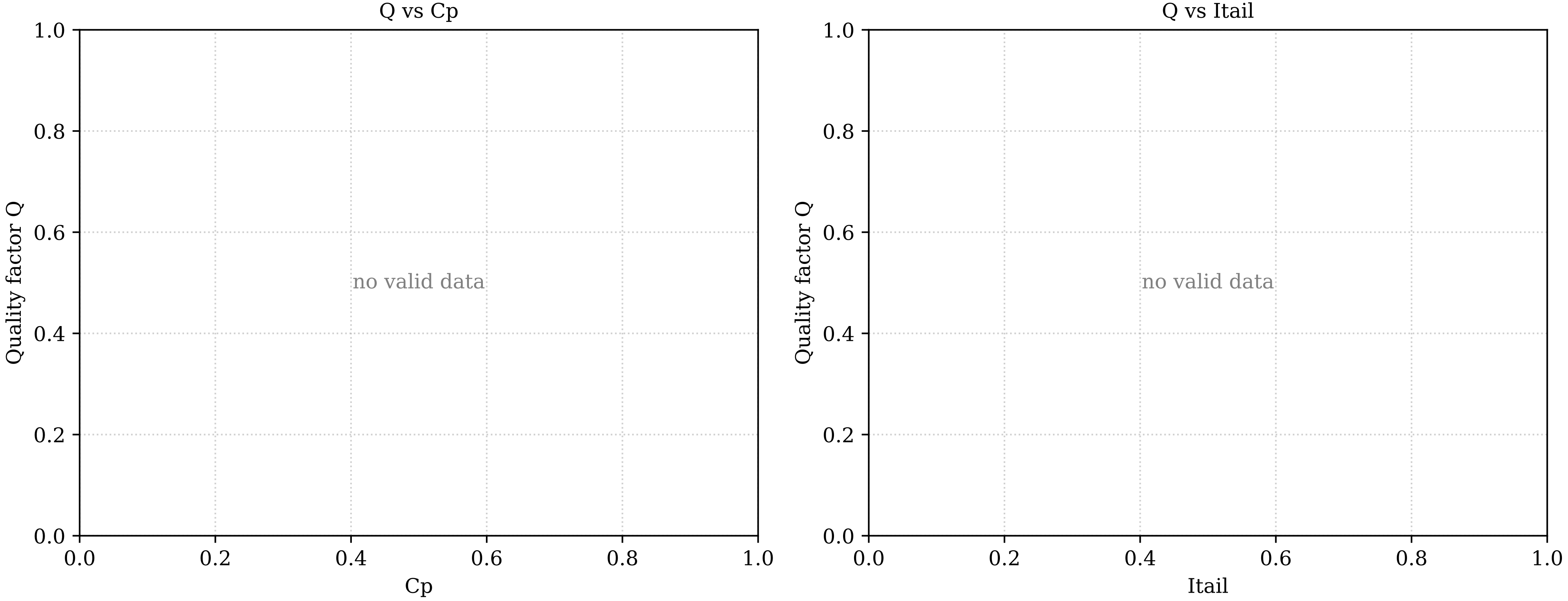
B. V(act+,act-) — Quality factor vs swept parameters



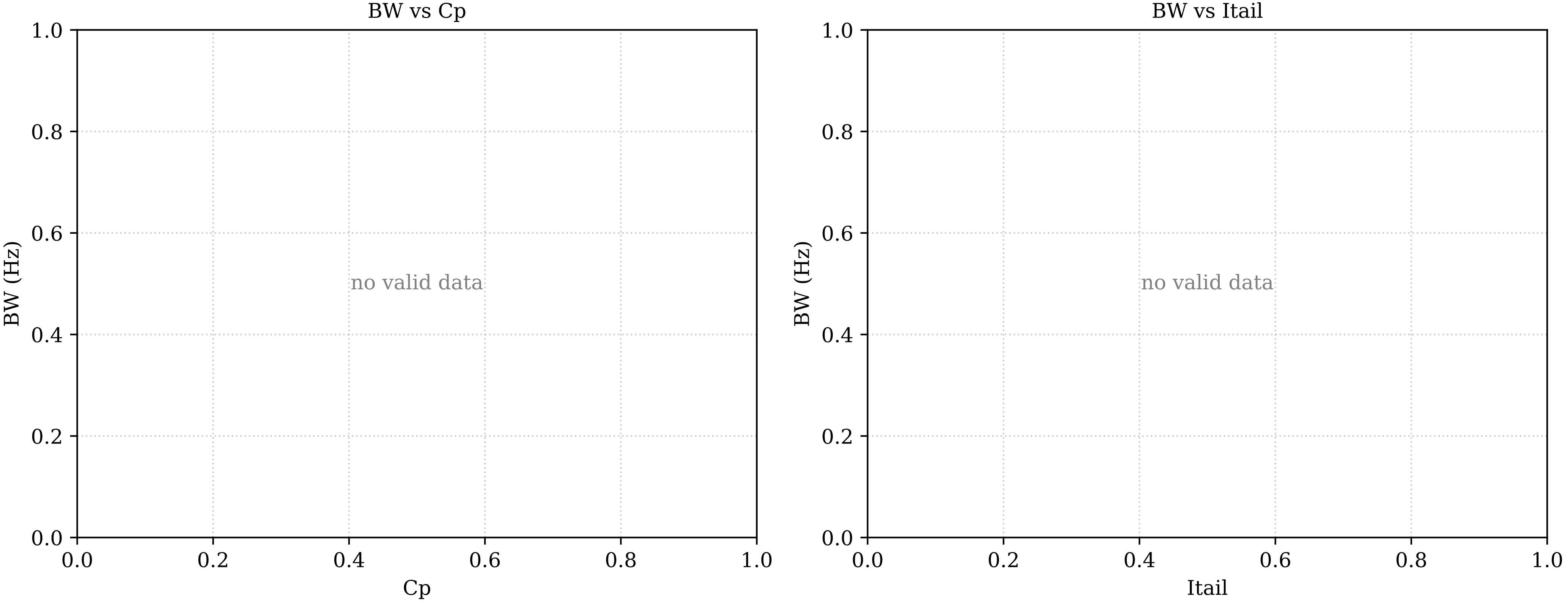
C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



B. V(pas+) — Quality factor vs swept parameters



C. V(pas+) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=0	Itail=10µ	Itail=20µ	Itail=30µ	Itail=40µ	Itail=50µ	Itail=60µ	Itail=70µ	Itail=80µ	Itail=90µ	Itail=100µ
Cp=400f	no BW	1135	364	304	285	282	288	302	328	371	452
Cp=450f	no BW	305	248	232	228	230	238	250	269	298	345
Cp=500f	1319	212	194	189	190	194	201	211	226	246	275

Red = BW could not be measured | Yellow = Q computed (Q ≤ 1400) | Green = Q > 1400

E. Combined quality-factor table — all signals (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	400f	0	91.25	-1.221	2.103			notch shallower than 3 d
V(act+,act-)	400f	10µ	91.26	-1.221	3.051	8.04e+04	1135	
V(act+,act-)	400f	20µ	91.26	-1.221	3.722	2.507e+05	364	
V(act+,act-)	400f	30µ	91.27	-1.221	4.466	3.002e+05	304	
V(act+,act-)	400f	40µ	91.27	-1.221	5.36	3.199e+05	285.3	
V(act+,act-)	400f	50µ	91.27	-1.221	6.506	3.238e+05	281.8	
V(act+,act-)	400f	60µ	91.27	-1.221	8.077	3.172e+05	287.8	
V(act+,act-)	400f	70µ	91.28	-1.221	10.45	3.018e+05	302.4	
V(act+,act-)	400f	80µ	91.28	-1.221	14.75	2.784e+05	327.8	
V(act+,act-)	400f	90µ	91.28	-1.221	29.13	2.461e+05	370.8	
V(act+,act-)	400f	100µ	91.28	-1.221	15.55	2.021e+05	451.7	
V(act+,act-)	450f	0	91.17	-1.316	2.553			notch shallower than 3 d
V(act+,act-)	450f	10µ	91.18	-1.315	3.662	2.992e+05	304.7	
V(act+,act-)	450f	20µ	91.19	-1.315	4.435	3.68e+05	247.8	
V(act+,act-)	450f	30µ	91.19	-1.315	5.279	3.932e+05	231.9	
V(act+,act-)	450f	40µ	91.2	-1.315	6.281	4.001e+05	227.9	
V(act+,act-)	450f	50µ	91.2	-1.315	7.546	3.959e+05	230.3	
V(act+,act-)	450f	60µ	91.2	-1.315	9.249	3.836e+05	237.7	
V(act+,act-)	450f	70µ	91.2	-1.315	11.77	3.645e+05	250.2	
V(act+,act-)	450f	80µ	91.2	-1.315	16.17	3.388e+05	269.2	
V(act+,act-)	450f	90µ	91.2	-1.315	30.02	3.06e+05	298.1	
V(act+,act-)	450f	100µ	91.2	-1.315	18.13	2.644e+05	344.9	
V(act+,act-)	500f	0	91.09	-1.417	3.017	6.909e+04	1319	
V(act+,act-)	500f	10µ	91.1	-1.417	4.28	4.307e+05	211.5	
V(act+,act-)	500f	20µ	91.11	-1.417	5.147	4.694e+05	194.1	
V(act+,act-)	500f	30µ	91.12	-1.417	6.082	4.809e+05	189.5	
V(act+,act-)	500f	40µ	91.12	-1.417	7.18	4.793e+05	190.1	
V(act+,act-)	500f	50µ	91.12	-1.417	8.545	4.693e+05	194.2	
V(act+,act-)	500f	60µ	91.12	-1.417	10.36	4.53e+05	201.2	
V(act+,act-)	500f	70µ	91.13	-1.417	12.98	4.311e+05	211.4	
V(act+,act-)	500f	80µ	91.13	-1.417	17.46	4.038e+05	225.7	
V(act+,act-)	500f	90µ	91.13	-1.417	30.78	3.709e+05	245.7	

E. Combined quality-factor table — all signals (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	500f	100µ	91.13	-1.417	20.41	3.312e+05	275.1	
V(pas+)	400f	0	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	10µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	20µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	30µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	40µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	50µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	60µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	70µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	80µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	90µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	400f	100µ	91.24	-6.146	0.6446			notch shallower than 3 d
V(pas+)	450f	0	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	10µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	20µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	30µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	40µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	50µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	60µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	70µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	80µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	90µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	450f	100µ	91.16	-6.178	0.8026			notch shallower than 3 d
V(pas+)	500f	0	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	10µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	20µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	30µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	40µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	50µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	60µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	70µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	80µ	91.08	-6.213	0.9736			notch shallower than 3 d

E. Combined quality-factor table — all signals (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	500f	90µ	91.08	-6.213	0.9736			notch shallower than 3 d
V(pas+)	500f	100µ	91.08	-6.213	0.9736			notch shallower than 3 d